RAG Learning

Mechanics

Kinematics (Motion in a Straight Line)

Key quantities:

- s = displacement (m)
- u = initial velocity (m/s)
- v = final velocity (m/s)
- a = acceleration (m/s^2)
- t = time (s)

SUVAT Equations (Constant Acceleration)

Equation	Missing Variable
$v = u + at$	s
$s = ut + \frac{1}{2}at^2$	v
$s = vt - \frac{1}{2}at^2$	u
$v^2 = u^2 + 2as$	t
$s = \frac{1}{2}(u + v)t$	a

Worked Example

A particle starts from rest and accelerates at 10 m/s^2 for 4 s. Find the distance travelled.

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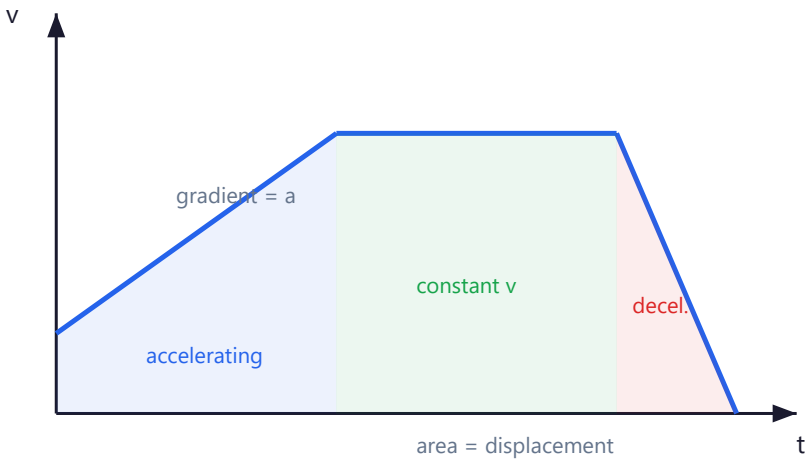
$u = 0$, $a = 10$, $t = 4$, find s:

$$s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(10)(16) = 80 \text{ m}$$

Velocity-Time Graphs

- **Gradient** = acceleration
- **Area under the graph** = displacement



Exam Tip: SUVAT only works for **constant acceleration**. If acceleration varies, use calculus: $v = \frac{ds}{dt}$, $a = \frac{dv}{dt}$, $s = \int v\,dt$.

Variable Acceleration

When acceleration is not constant, use calculus:

$$v = \frac{ds}{dt} \qquad a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

$$s = \int v\,dt \qquad v = \int a\,dt$$

Worked Example: Differentiation

A particle has velocity $\mathbf{v} = (3t^2)\mathbf{i} + (2t)\mathbf{j}$. Find the acceleration at $t = 2$.

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = (6t)\mathbf{i} + 2\mathbf{j}$$

$$\text{At } t = 2: \quad \mathbf{a} = 12\mathbf{i} + 2\mathbf{j}$$

Worked Example: Integration to Find Displacement

A particle has acceleration $a = (6t - 2)$ m/s^2. At $t = 0$, the velocity is 3 m/s and displacement is 0. Find the displacement when $t = 3$.

Step 1: Find velocity

$$v = \int (6t - 2)\, dt = 3t^2 - 2t + c$$

At $t = 0$, $v = 3$: $c = 3$. So $v = 3t^2 - 2t + 3$.

Step 2: Find displacement

$$s = \int (3t^2 - 2t + 3)\, dt = t^3 - t^2 + 3t + k$$

At $t = 0$, $s = 0$: $k = 0$. At $t = 3$:

$$s = 27 - 9 + 9 = 27 \text{ m}$$

Forces and Newton's Laws

Newton's Three Laws

1. An object remains at rest or in uniform motion unless acted on by a resultant force
2. $\mathbf{F} = m\mathbf{a}$ — the resultant force equals mass times acceleration
3. Every action has an equal and opposite reaction

Common Forces

Force	Symbol	Direction
Weight	$W = mg$	Vertically downward
Normal reaction	R or N	Perpendicular to surface
Friction	$F = \mu R$	Opposite to motion/tendency
Tension	T	Along the string, away from object

Worked Example

A 5 kg block on a rough surface ($\mu = 0.4$) is pulled with 30 N horizontally. Find the acceleration.

Normal reaction: $R = mg = 5 \times 9.8 = 49$ N

Friction: $F = \mu R = 0.4 \times 49 = 19.6$ N

$$a = \frac{30 - 19.6}{5} = \frac{10.4}{5} = 2.08 \text{ m/s}^2$$

Connected Particles

Pulleys

For two particles of mass m_1 and m_2 connected by a string over a smooth pulley (with $m_1 > m_2$):

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2} \qquad T = \frac{2m_1 m_2 g}{m_1 + m_2}$$

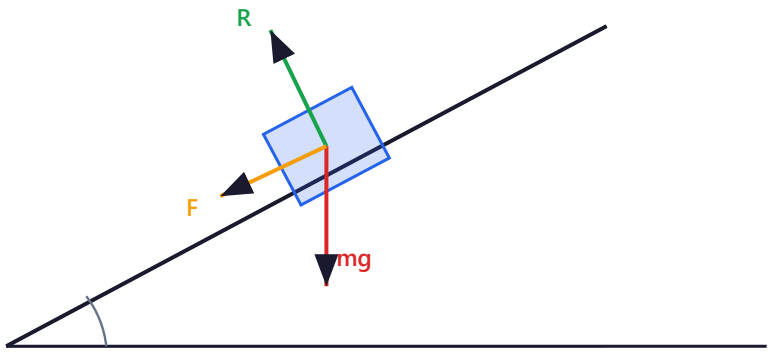
Worked Example

Masses of 3 kg and 2 kg are connected over a smooth pulley. Find the acceleration ($g = 10$).

$$a = \frac{(3 - 2) \times 10}{3 + 2} = \frac{10}{5} = 2 \text{ m/s}^2$$

Exam Tip: For connected particle problems, draw a **separate force diagram** for each particle and apply $F = ma$ to each. The string tension is the same throughout (for a light, inextensible string).

Motion on Inclined Planes



Resolve forces parallel and perpendicular to the slope:

- **Along the plane:** $mg \sin \alpha$ (down the slope)
- **Perpendicular:** Normal reaction $R = mg \cos \alpha$
- **Friction:** $F = \mu R = \mu mg \cos \alpha$

Worked Example

A 3 kg object slides down a smooth incline at 30°. Find its acceleration.

$$a = g \sin 30^\circ = 9.8 \times 0.5 = 4.9 \text{ m/s}^2$$

Projectiles

A projectile launched at speed u at angle θ to the horizontal:

- **Horizontal:** constant velocity $u \cos \theta$ (no horizontal acceleration)
- **Vertical:** initial velocity $u \sin \theta$, acceleration $-g$

Key Formulae

$$\text{Time of flight} = \frac{2u\sin\theta}{g}$$

$$\text{Maximum height} = \frac{u^2\sin^2\theta}{2g}$$

$$\text{Range} = \frac{u^2\sin 2\theta}{g}$$

Worked Example: Full Projectile Problem

A ball is projected from ground level at 30° to the horizontal with speed 20 m/s. Find: (a) the time of flight, (b) the maximum height, (c) the range. ($g = 10$ m/s^2)

Horizontal: $v_x = 20\cos 30^\circ = 10\sqrt{3}$ m/s (constant)

Vertical: $u_y = 20\sin 30^\circ = 10$ m/s, $a = -10$ m/s^2

(a) Time of flight: Land when $s_y = 0$:

$$s_y = u_y t + \tfrac{1}{2}at^2 = 10t - 5t^2 = 0 \implies t(10-5t) = 0 \implies t = 2\text{ s}$$

(b) Maximum height: At top, $v_y = 0$:

$$v_y^2 = u_y^2 + 2as \implies 0 = 100 - 20H \implies H = 5\text{ m}$$

(c) Range:

$$R = v_x \times t = 10\sqrt{3} \times 2 = 20\sqrt{3} \approx 34.6 \text{ m}$$

Exam Tip: Treat horizontal and vertical motion separately. The horizontal velocity is constant (no air resistance). Use SUVAT vertically with $a = -g$ (taking upward as positive).

Moments

Moment of a force about a point = force $\times$ perpendicular distance from the point to the line of action.

$$M = F \times d$$

For equilibrium:

- Resultant force = 0
- Sum of clockwise moments = sum of anticlockwise moments (about any point)

Worked Example

A uniform beam of length 6 m and weight 120 N rests on supports at its ends A and B. A 60 N weight is placed 2 m from A. Find the reaction at A.

Taking moments about B:

$$R_A \times 6 = 120 \times 3 + 60 \times 4 = 360 + 240 = 600$$

$$R_A = 100 \text{ N}$$

Check: $R_B = 120 + 60 - 100 = 80$ N.

Momentum and Impulse

Momentum: $p = mv$ (vector quantity, units: kg m/s or N s)
Impulse: $I = Ft = mv - mu$ (change in momentum)
Conservation of momentum: In a closed system, total momentum before = total momentum after.

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Coefficient of Restitution

$$e = \frac{\text{speed of separation}}{\text{speed of approach}} \qquad 0 \leq e \leq 1$$

$e = 1$: perfectly elastic. $e = 0$: perfectly inelastic (objects coalesce).

Worked Example

A 0.5 kg ball moving at 4 m/s strikes a wall and rebounds at 3 m/s. Find the impulse.

Taking rebound direction as positive:

$$I = mv - mu = 0.5(3) - 0.5(-4) = 1.5 + 2 = 3.5 \text{ N s}$$

Work, Energy, and Power

Work done: $W = Fs\cos\theta$ (force $\times$ displacement $\times$ cosine of angle between them)
Kinetic energy: $KE = \frac{1}{2}mv^2$
Gravitational PE: $GPE = mgh$
Power: $P = \frac{W}{t} = Fv$

Work-Energy Principle

Net work done on a body = change in kinetic energy:

$$W_{\text{net}} = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

Conservation of Energy

Total energy is conserved (KE + PE + work done against friction):

$$\frac{1}{2}mu^2 + mgh_1 = \frac{1}{2}mv^2 + mgh_2 + F_{\text{friction}} \times d$$

Worked Example

A 2 kg block slides 5 m down a rough incline (30°, $\mu = 0.3$). Find its speed at the bottom (starting from rest).

Height lost: $h = 5\sin 30^\circ = 2.5$ m

Friction: $F = \mu mg\cos 30^\circ = 0.3 \times 2 \times 9.8 \times \cos 30^\circ \approx 5.09$ N

$$\frac{1}{2}(2)v^2 = 2(9.8)(2.5) - 5.09(5)$$

$$v^2 = 49 - 25.45 = 23.55 \implies v \approx 4.85 \text{ m/s}$$

Exam Tip: In energy problems, always draw a clear diagram and define your reference level for GPE. The work-energy approach often avoids the need to find acceleration — useful when the force varies or the path is complex.

Resolving Forces

Any force can be resolved into two perpendicular components:

$$F_x = F\cos\theta \qquad F_y = F\sin\theta$$

The resultant of two perpendicular forces F_x and F_y:

$$R = \sqrt{F_x^2 + F_y^2} \qquad \tan\alpha = \frac{F_y}{F_x}$$

Worked Example

Two forces of 5 N and 12 N act perpendicular to each other. Find the resultant.

$$R = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13 \text{ N}$$